

NCERT Solutions for Class 12 Maths

Chapter 11 – Three-Dimensional Geometry

Exercise 11.1

1.If a line makes angles $90^\circ, 135^\circ, 45^\circ$ with x, y and z axes respectively, find its direction cosines.

Ans: Let us consider l, m and n be the direction cosines of line.

Then,

$$l = \cos 90^\circ = 0,$$

$$m = \cos 135^\circ$$

$$= \cos(90^\circ + 45^\circ)$$

$$= \sin 45^\circ$$

$$= -\frac{1}{\sqrt{2}}$$

And,

$$n = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Therefore, the direction cosines of the line are $0, -\frac{1}{\sqrt{2}}$ and $\frac{1}{\sqrt{2}}$.

2. Find the direction cosines of the line which makes equal angles with the coordinate axes.

Ans: Let us consider that the line makes an angle α with coordinate axes.

Which means $l = \cos \alpha, m = \cos \alpha, n = \cos \alpha$

Now, we know that

$$l^2 + m^2 + n^2 \Rightarrow \cos^2 \alpha + \cos^2 \alpha + \cos^2 \alpha$$

$$\Rightarrow 3\cos^2 \alpha = 1$$

$$\Rightarrow \cos^2 \alpha = \frac{1}{3} \Rightarrow \cos \alpha = \pm \frac{1}{\sqrt{3}}$$

Therefore, the direction cosines of the line are $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$.

3.If a line has the direction ratios -18,12,-4, then what are its direction cosines?

Ans: We have the direction ratios as -18,12,-4,

Now, the direction cosines will be as

$$l = \frac{-18}{\sqrt{(-18)^2 + (12)^2 + (-4)^2}}, m = \frac{12}{\sqrt{(-18)^2 + (12)^2 + (-4)^2}}, n = \frac{-4}{\sqrt{(-18)^2 + (12)^2 + (-4)^2}}$$

$$\frac{-18}{22}, \frac{12}{22}, \frac{-4}{22} \Rightarrow \frac{-9}{11}, \frac{6}{11}, \frac{-2}{11}$$

Therefore, direction cosines of the line are $\frac{-9}{11}, \frac{6}{11}$ and $\frac{-2}{11}$.

4. Show that the points $(2, 3, 4), (-1, -2, 1), (5, 8, 7)$ are collinear.

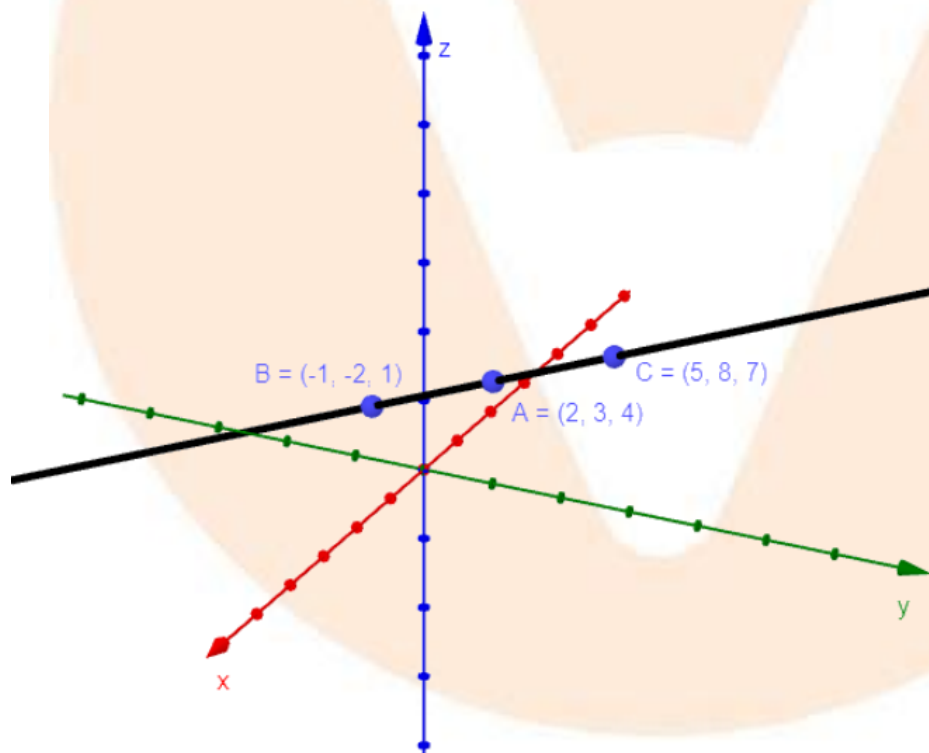
Ans: Let us consider the points be $A(2, 3, 4), B(-1, -2, 1)$ and $C(5, 8, 7)$.

Now, as we know that direction cosines can be found by $(x_2 - x_1), (y_2 - y_1)$, and $(z_2 - z_1)$

Therefore,

Direction ratios of AB and BC be $-3, -5, -3$ and $6, 10, 6$ respectively

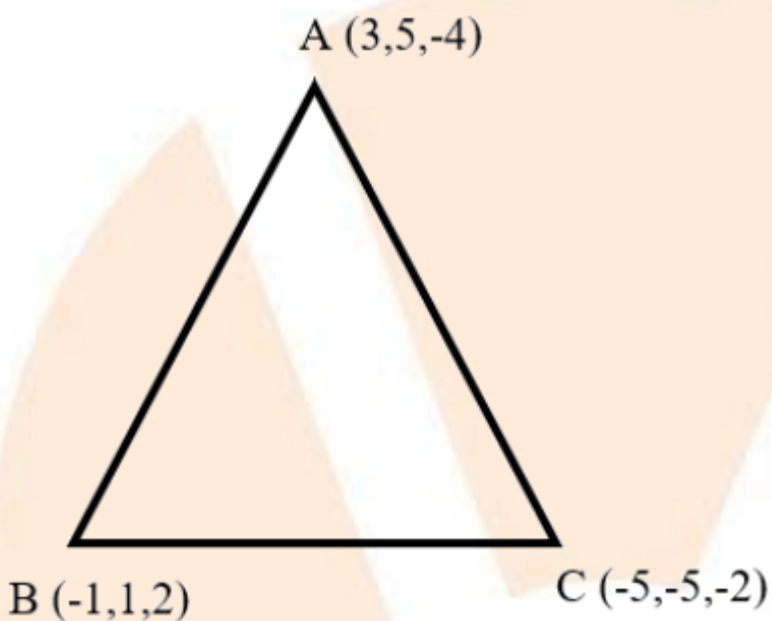
As we can see that AB and BC are proportional, we get that AB is parallel to BC .



Therefore, the points are collinear.

5. Find the direction cosines of the sides of the triangle whose vertices are $(3, 5, -4), (-1, 1, 2), (-5, -5, -2)$.

Ans: Let us consider the points be $A(3,5,-4)$, $B(-1,1,2)$ and $C(-5,-5,-2)$,



Now, the direction ratios of AB will be $-4, -4$ and 6 ,

We get

$$\sqrt{(-4)^2 + (-4)^2 + (6)^2} = \sqrt{68} \Rightarrow 2\sqrt{17}$$

Now,

$$l = \frac{-4}{\sqrt{(-4)^2 + (-4)^2 + (6)^2}}, m = \frac{-4}{\sqrt{(-4)^2 + (-4)^2 + (6)^2}}, n = \frac{6}{\sqrt{(-4)^2 + (-4)^2 + (6)^2}}$$

$$\Rightarrow l = \frac{-2}{\sqrt{17}}, m = \frac{-2}{\sqrt{17}}, n = \frac{3}{\sqrt{17}}$$

Therefore, the direction cosines of AB are $\frac{-2}{\sqrt{17}}, \frac{-2}{\sqrt{17}}, \frac{3}{\sqrt{17}}$

Similarly, the direction ratios of side BC will be $-4, -6$ and -4 .

Now,

$$l = \frac{-4}{\sqrt{(-4)^2 + (-6)^2 + (-4)^2}}, m = \frac{-6}{\sqrt{(-4)^2 + (-6)^2 + (-4)^2}}, n = \frac{-4}{\sqrt{(-4)^2 + (-6)^2 + (-4)^2}}$$

$$l = \frac{-4}{2\sqrt{17}}, m = \frac{-6}{2\sqrt{17}}, n = \frac{-4}{2\sqrt{17}}$$

Therefore, the direction cosines of BC is $\frac{-2}{\sqrt{17}}, \frac{-3}{\sqrt{17}}, \frac{-2}{\sqrt{17}}$

Similarly, the direction ratios of CA will be $-8, -10$ and 2 .

Now,

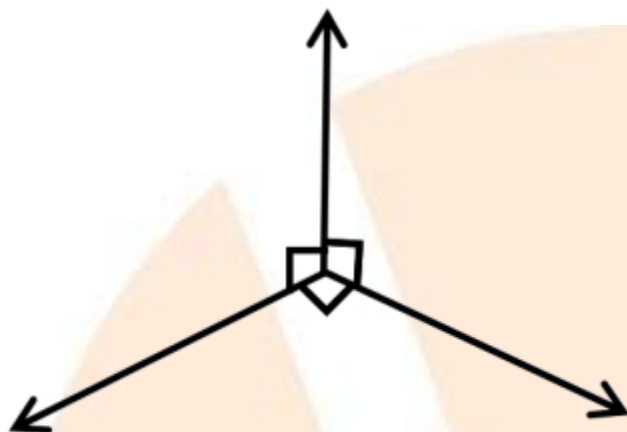
$$l = \frac{-8}{\sqrt{(-8)^2 + (-10)^2 + (2)^2}}, m = \frac{-10}{\sqrt{(-8)^2 + (-10)^2 + (2)^2}}, n = \frac{2}{\sqrt{(-8)^2 + (-10)^2 + (2)^2}}$$

$$l = \frac{-8}{2\sqrt{42}}, m = \frac{-10}{2\sqrt{42}}, n = \frac{2}{2\sqrt{42}}.$$

Therefore, the direction cosines of CA is $\frac{-4}{\sqrt{42}}, \frac{-5}{\sqrt{42}}, \frac{1}{\sqrt{42}}$

Exercise 11.2

1. Show that the three lines with direction cosines $\frac{12}{13}, \frac{-3}{13}, \frac{-4}{13}; \frac{4}{13}, \frac{12}{13}, \frac{3}{13}; \frac{3}{13}, \frac{-4}{13}, \frac{12}{13}$ are mutually perpendicular,



Ans: As we know, if $l_1l_2 + m_1m_2 + n_1n_2 = 0$, the lines are perpendicular.

(i) Now, from direction cosines $\frac{12}{13}, \frac{-3}{13}, \frac{-4}{13}$ and $\frac{4}{13}, \frac{12}{13}, \frac{3}{13}$, we get

$$l_1l_2 + m_1m_2 + n_1n_2 = \frac{12}{13} \times \frac{4}{13} + \left(\frac{-3}{13}\right) \times \frac{12}{13} + \left(\frac{-4}{13}\right) \times \frac{3}{13}$$

$$\Rightarrow \frac{48}{169} - \frac{36}{169} - \frac{12}{169}$$

$$\Rightarrow 0$$

Therefore, the lines are perpendicular.

(ii) Similarly, if we take $\frac{4}{13}, \frac{12}{13}, \frac{3}{13}$ and $\frac{12}{13}, \frac{-3}{13}, \frac{-4}{13}$, we get

$$l_1l_2 + m_1m_2 + n_1n_2 = \frac{4}{13} \times \frac{3}{13} + \left(\frac{12}{13}\right) \times \left(\frac{-4}{13}\right) + \frac{3}{13} \times \left(\frac{12}{13}\right)$$

$$\Rightarrow \frac{12}{169} - \frac{48}{169} - \frac{36}{169} = 0$$

Therefore, the lines are perpendicular.

(iii) Again, if we consider $\frac{-3}{13}, \frac{-4}{13}, \frac{12}{13}$ and $\frac{12}{13}, \frac{-3}{13}, \frac{-4}{13}$, we get

$$l_1l_2 + m_1m_2 + n_1n_2 = \frac{3}{13} \times \frac{12}{13} + \left(\frac{-4}{13}\right) \times \left(\frac{-4}{13}\right) + \frac{12}{13} \times \left(\frac{-4}{13}\right)$$

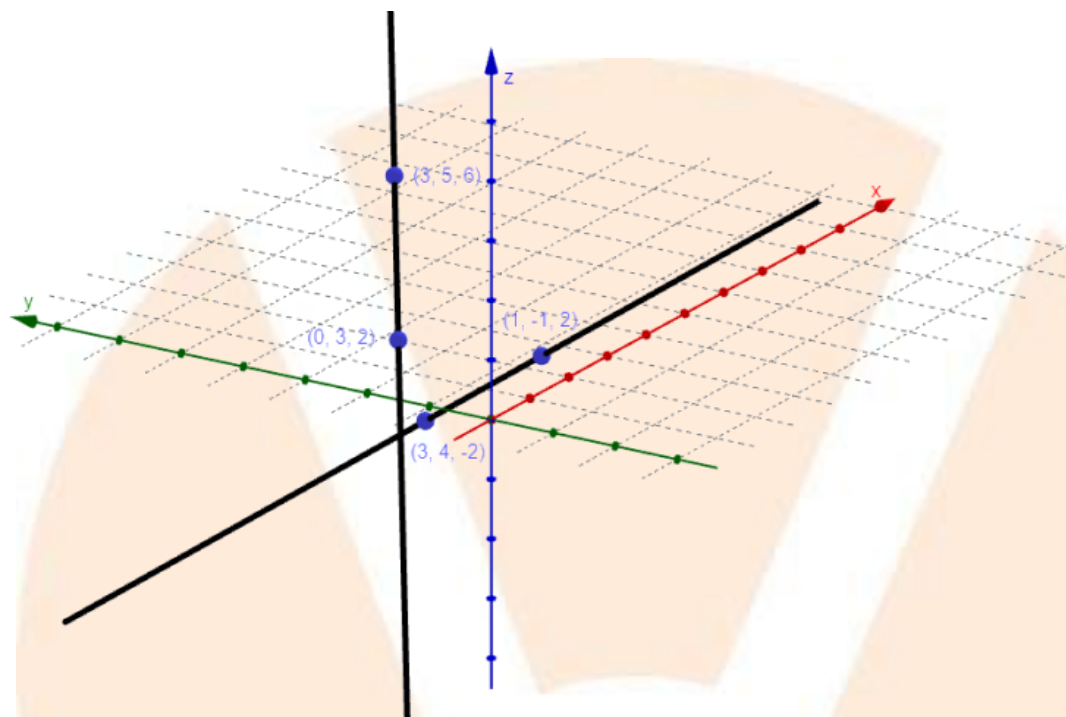
$$\Rightarrow \frac{36}{169} - \frac{12}{169} - \frac{48}{169} = 0$$

Therefore, the lines are perpendicular.

Therefore, we can say that all the lines are mutually perpendicular.

2. show that the line passing through the points $(1, -1, 2)$ and $(3, 4, -2)$ is perpendicular to the line through the points $(0, 3, 2)$ and $(3, 5, 6)$?

Ans: Let us consider that AB and CD are the lines that pass through the points, $(1, -1, 2)$, $(3, 4, -2)$ and $(0, 3, 2)$, $(3, 5, 6)$, respectively.



Now, we have $a_1 = (2), b_1 = (5), c_1 = (-4)$ and $a_2 = (3), b_2 = (2), c_2 = (4)$

As we know that if $AB \perp CD$ then $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Now,

$$a_1a_2 + b_1b_2 + c_1c_2 = 2 \times 3 + 5 \times 2 + (-4) \times 4$$

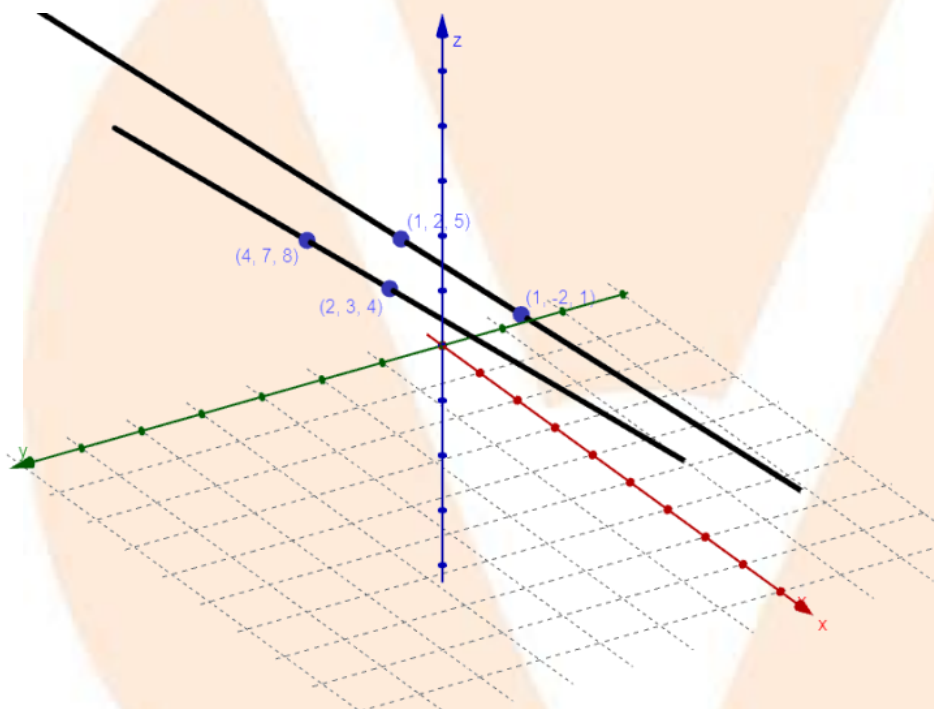
$$\Rightarrow 2 \times 3 + 5 \times 2 - 4 \times 4 = 6 + 10 - 16$$

$$\Rightarrow 0$$

Therefore, AB and CD are perpendicular to each other.

3. Show that the line through the points $(4, 7, 8)$ and $(2, 3, 4)$ is parallel to the line through the points $(1, -2, 1)$ and $(1, 2, 5)$.

Ans: Let us consider the lines AB and CD that pass through points $(4, 7, 8)$, $(2, 3, 4)$, and $(1, -2, 1)$, $(1, 2, 5)$ respectively.



Now, we get

$$a_1 = (2 - 4), b_1 = (3 - 7), c_1 = (4 - 8) \text{ and } a_2 = (1 + 1), b_2 = (2 + 2), c_2 = (5 - 1)$$

$$a_1 = (-2), b_1 = (-4), c_1 = (-4) \text{ and } a_2 = (2), b_2 = (4), c_2 = (4)$$

Now, we know that if $AB \parallel CD$ then $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$,

Now,

$$\frac{a_1}{a_2} = \frac{-2}{2} \Rightarrow -1, \frac{b_1}{b_2} = \frac{-4}{4} \Rightarrow -1, \frac{c_1}{c_2} = \frac{-4}{4} = -1$$

We got $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Therefore, AB is parallel to CD .

4. Find the equation of the line which passes through point $(1, 2, 3)$ and parallel to the vector $3\hat{i} + 2\hat{j} - 2\hat{k}$.

Ans: Now, let us consider the position vector A be $a = \hat{i} + 2\hat{j} + 3\hat{k}$ and let $\vec{b} = 3\hat{i} + 2\hat{j} - 2\hat{k}$

Now, we know that the line passes through A and is parallel to $\vec{b}$,

As we know $\vec{r} = \vec{a} + \lambda\vec{b}$ where λ is a constant

$$\Rightarrow \vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(3\hat{i} + 2\hat{j} - 2\hat{k})$$

Therefore, the equation of the line is $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(3\hat{i} + 2\hat{j} - 2\hat{k})$

5. Find the equation of the line in vector and in Cartesian form that passes through the point with position vector $2\hat{i} - \hat{j} + 4\hat{k}$ and is in the direction $\hat{i} + 2\hat{j} - \hat{k}$.

Ans: We know that the line passes through the point with position vector.

Now, let us consider $\vec{a} = 2\hat{i} - \hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$

Now, line passes through point A and parallel to $\vec{b}$, we get.

$$\vec{r} = 2\hat{i} - \hat{j} + 4\hat{k} + \lambda(\hat{i} + 2\hat{j} - \hat{k})$$

Therefore, the equation of the line in vector form is $\vec{r} = 2\hat{i} - \hat{j} + 4\hat{k} + \lambda(\hat{i} + 2\hat{j} - \hat{k})$.

Now, we know

$$\vec{r} = x\hat{i} - y\hat{j} + z\hat{k} \Rightarrow x\hat{i} - y\hat{j} + z\hat{k} = (\lambda + 2)\hat{i} + (2\lambda - 1)\hat{j} + (-\lambda + 4)\hat{k}$$

Therefore, the equation of the line in cartesian form will be $\frac{x-2}{1} = \frac{y+1}{2} = \frac{z-4}{-1}$.

6.find the Cartesian equation of the line which passes through the point $(-2, 4, -5)$ and parallel to the line given by $\frac{x+3}{3} = \frac{y-4}{5} = \frac{z+8}{6}$.

Ans: We know that the line passes through point $(-2, 4, -5)$ and also parallel to $\frac{x+3}{3} = \frac{y-4}{5} = \frac{z+8}{6}$

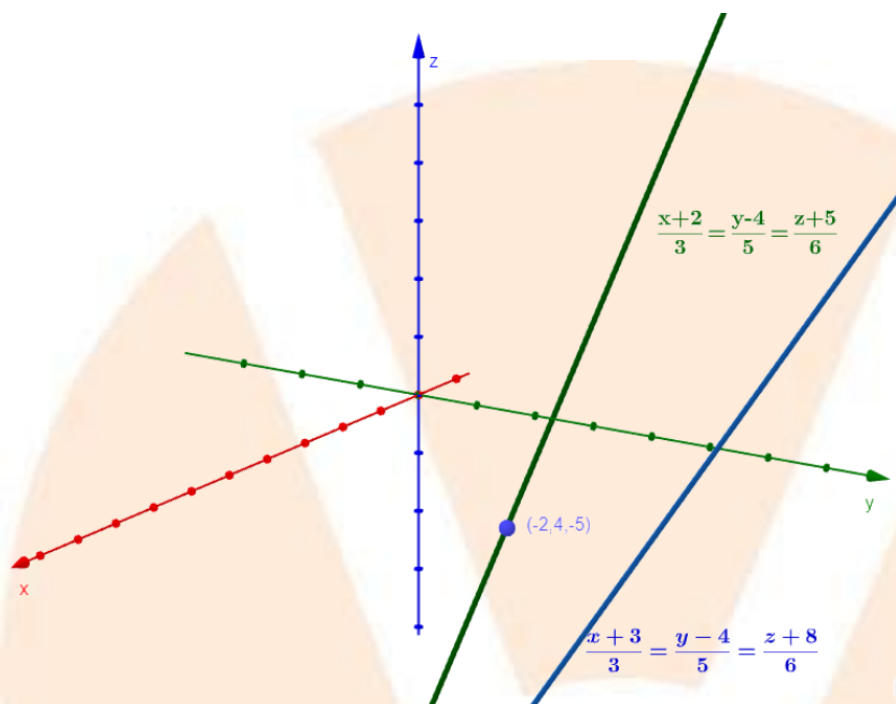
Now, as we can see the direction ratios of the line are 3, 5 and 6.

As we know the required line is parallel to $\frac{x+3}{3} = \frac{y-4}{5} = \frac{z+8}{6}$

Therefore, the direction ratios will be $3k, 5k$ and $6k$

As we know that the equation of the line through the point and with direction ratio is shown in form $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

Therefore, the equation of the line $\frac{x+2}{3} = \frac{y-4}{5} = \frac{z+5}{6} = k$.



7. The Cartesian equation of a line is $\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$ Write its vector form.

Ans: As we can see the cartesian equation of the line, we can tell that the line is passing through $(5, -4, 6)$, and the direction ratios are 3, 7 and 2.

Now, we got the position vector $\vec{a} = 5\hat{i} - 4\hat{j} + 6\hat{k}$

From this we got the direction of the vector be $\vec{b} = 3\hat{i} + 7\hat{j} + 2\hat{k}$

Therefore, the vector form of the line will be $\vec{r} = 5\hat{i} - 4\hat{j} + 6\hat{k} + \lambda(3\hat{i} + 7\hat{j} + 2\hat{k})$

8. Find the angle between the following pair of lines

(i) $\vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k})$ and $\vec{r} = 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$

(ii) $\vec{r} = 3\hat{i} + \hat{j} - 2\hat{k} + \lambda(\hat{i} - \hat{j} - 2\hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} - 56\hat{k} + \mu(3\hat{i} - 5\hat{j} - 4\hat{k})$

(i) Ans: (i) Let us consider the angle be θ ,

As we know that the angle between the lines can be found by $\cos\theta = \frac{|\vec{b}_1 \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|}$

As the lines are parallel to $\vec{b}_1 = 3\hat{i} + 2\hat{j} + 6\hat{k}$ and $\vec{b}_2 = \hat{i} + 2\hat{j} + 2\hat{k}$, we got.

$$|\vec{b}_1| = \sqrt{3^2 + 2^2 + 6^2} = 7, |\vec{b}_2| = \sqrt{1^2 + 2^2 + 2^2} = 3 \text{ and } \vec{b}_1 \vec{b}_2 = (3\hat{i} + 2\hat{j} + 6\hat{k})(\hat{i} + 2\hat{j} + 2\hat{k}) = 19$$

Therefore, the angle between the lines will be

$$\cos\theta = \frac{19}{7 \times 3}$$

$$\Rightarrow \theta = \cos^{-1} \frac{19}{21}$$

(ii) $\vec{r} = 3\hat{i} + \hat{j} - 2\hat{k} + \lambda(\hat{i} - \hat{j} - 2\hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} - 56\hat{k} + \mu(3\hat{i} - 5\hat{j} - 4\hat{k})$

As the lines are parallel to the vectors $\vec{b}_1 = \hat{i} - \hat{j} - 2\hat{k}$ and $\vec{b}_2 = 3\hat{i} - 5\hat{j} - 4\hat{k}$, we get

$$|\vec{b}_1| = \sqrt{1^2 + (-1)^2 + (-2)^2} = \sqrt{6}, \quad |\vec{b}_2| = \sqrt{3^2 + (-5)^2 + (-4)^2} = 5\sqrt{2} \quad \text{and}$$

$$\vec{b}_1 \vec{b}_2 = (\hat{i} - \hat{j} - 2\hat{k})(3\hat{i} - 5\hat{j} - 4\hat{k}) = 16$$

Therefore, the angle between them will be,

$$\cos\theta = \frac{16}{10\sqrt{3}}$$

$$\Rightarrow \cos\theta = \frac{8}{5\sqrt{3}}$$

$$\Rightarrow \theta = \cos^{-1} \frac{8}{5\sqrt{3}}$$

9. Find the angle between the following pair of lines

$$(i) \frac{x-2}{2} = \frac{y-1}{5} = \frac{z+3}{-3} \text{ and } \frac{x+2}{-1} = \frac{y-4}{8} = \frac{z-5}{4}$$

$$(ii) \frac{x}{2} = \frac{y}{2} = \frac{z}{1} \text{ and } \frac{x-5}{4} = \frac{y-2}{1} = \frac{z-3}{8}$$

Ans: (i) Let us take $\vec{b}_1$ and $\vec{b}_2$ be the vectors parallel to the lines, we get

$$\vec{b}_1 = 2\hat{i} + 5\hat{j} - 3\hat{k} \text{ and } \vec{b}_2 = -\hat{i} + 8\hat{j} + 4\hat{k}$$

$$\text{Now } |\vec{b}_1| = \sqrt{2^2 + 5^2 + (-3)^2} = \sqrt{38}, |\vec{b}_2| = \sqrt{(-1)^2 + 8^2 + 4^2} = 9$$

And,

$$\vec{b}_1 \vec{b}_2 = (2\hat{i} + 5\hat{j} - 3\hat{k})(-\hat{i} + 8\hat{j} + 4\hat{k})$$

$$= 2(-1) + 5(8) + 4(-3)$$

$$= 26$$

We can find the angle by using $\cos\theta = \frac{|\vec{b}_1 \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|}$

Therefore,

$$\cos\theta = \frac{26}{9\sqrt{38}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$$

Therefore, the angle will be $\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$.

(ii)

Ans: similarly let us consider $\vec{b}_1$ and $\vec{b}_2$ be the vectors parallel to lines, we get

$$\vec{b}_1 = 2\hat{i} + 2\hat{j} + \hat{k} \text{ and } \vec{b}_2 = 4\hat{i} + \hat{j} + 8\hat{k}$$

Now, $|\vec{b}_1| = \sqrt{2^2 + 2^2 + 1^2} = 3$, $|\vec{b}_2| = \sqrt{4^2 + 1^2 + 8^2} = 9$ and

$$\vec{b}_1 \vec{b}_2 = (2\hat{i} + 2\hat{j} + \hat{k}) \cdot (4\hat{i} + \hat{j} + 8\hat{k})$$

$$= 2(4) + 2(1) + 1(8)$$

$$= 18$$

As we know the angle can be found by $\cos\theta = \frac{|\vec{b}_1 \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|}$

Therefore,

$$\cos\theta = \frac{18}{27} = \frac{2}{3}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{2}{3}\right)$$

Therefore, the angle is $\cos^{-1}\left(\frac{2}{3}\right)$.

10. Find the values of p so the line $\frac{1-x}{3} = \frac{7y-14}{2p} = \frac{z-3}{2}$ and $\frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{2}$ are at right angles.

Ans: As we know that the correct form of the equation is as follows,

$$\frac{x-1}{-3} = \frac{y-2}{\frac{2p}{7}} = \frac{z-3}{2} \text{ and } \frac{x-1}{\frac{-3p}{7}} = \frac{y-5}{1} = \frac{z-6}{-5}$$

From this we get the direction ratios as

$$a_1 = -3, b_1 = \frac{2p}{7}, c_1 = 2 \text{ and } a_2 = \frac{-3p}{7}, b_2 = 1, c_2 = -5$$

As we know the lines are perpendicular, we get

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

$$\Rightarrow \frac{9p}{7} + \frac{2p}{7} = 10$$

$$\Rightarrow 11p = 70$$

$$\Rightarrow p = \frac{70}{11}$$

Therefore, the value of p is $\frac{70}{11}$.

11: show that the lines $\frac{x-5}{7} = \frac{y+2}{-5} = \frac{z}{1}$ and $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ are perpendicular to each other.

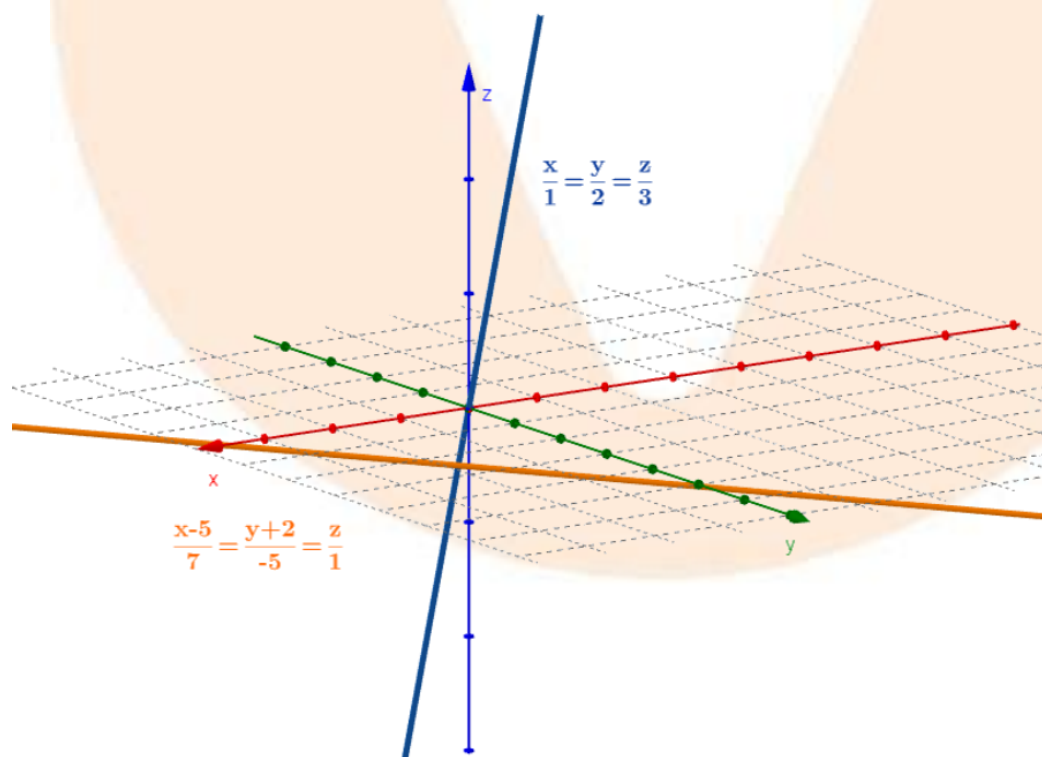
Ans: From the given equation, we get the direction ratios as,

$$a_1 = 7, b_1 = -5, c_1 = 1, a_2 = 1, b_2 = 2, c_2 = 3$$

As we know, if $a_1a_2 + b_1b_2 + c_1c_2 = 0$, the lines are perpendicular to each other

Now,

$$7(1) + (-5)(2) + 1(3) \Rightarrow 7 - 10 + 3 = 0$$



Therefore, the lines are perpendicular.

12. find the shortest distance between the lines are $\vec{r} = \hat{i} + 2\hat{j} + \hat{k} + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \mu(2\hat{i} + \hat{j} + 2\hat{k})$.

Ans: We have been given lines, $\vec{r} = \hat{i} + 2\hat{j} + \hat{k} + \lambda(\hat{i} - \hat{j} - \hat{k})$ and

$$\vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \mu(2\hat{i} + \hat{j} + 2\hat{k})$$

As we know that the shortest distance can be found as $d = \frac{|(\vec{b}_1 \times \vec{b}_2)(\vec{a}_2 - \vec{a}_1)|}{|\vec{b}_1 \times \vec{b}_2|}$

Now, from the given lines we get that

$$\vec{a}_1 = \hat{i} + 2\hat{j} + \hat{k},$$

$$\vec{b}_1 = \hat{i} - \hat{j} - \hat{k}$$

$$\vec{a}_2 = 2\hat{i} - \hat{j} - \hat{k},$$

$$\vec{b}_2 = 2\hat{i} + \hat{j} + 2\hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = (2\hat{i} - \hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + \hat{k})$$

$$= \hat{i} - 3\hat{j} - 2\hat{k},$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & -1 & 2 \end{vmatrix}$$

$$\Rightarrow \vec{b}_1 \times \vec{b}_2 = -3\hat{i} + 3\hat{k}.$$

$$\text{Then, } |\vec{b}_1 \times \vec{b}_2| = \sqrt{(-3)^2 + 3^2} = 3\sqrt{2}$$

Now, if we put all the values in their places, we get

$$d = \left| \frac{(-3\hat{i} + 3\hat{k})(\hat{i} - 3\hat{j} - 2\hat{k})}{3\sqrt{2}} \right| \Rightarrow d = \left| \frac{-3(1) + 3(2)}{3\sqrt{2}} \right|$$

$$d = \left| \frac{-9}{3\sqrt{2}} \right| \Rightarrow d = \frac{3\sqrt{2}}{2}$$

Therefore, the shortest distance between the lines is $\frac{3\sqrt{2}}{2}$ units.

13. Find the shortest distance between the lines

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \quad \text{and} \quad \frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$$

Ans: As we know that the shortest distance can be found by,

$$d = \frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2}}$$

Now, from the given lines we got that.

$$x_1 = -1, y_1 = -1, z_1 = -1, a_1 = 7, b_1 = -6, c_1 = 1$$

$$x_2 = 3, y_2 = 5, z_2 = 7, a_2 = 1, b_2 = -2, c_2 = 1$$

And,

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = \begin{vmatrix} 4 & 6 & 8 \\ 7 & -6 & 1 \\ 1 & -2 & 1 \end{vmatrix}$$

$$= 4(-6 + 2) - 6(1 + 7) + 8(-14 + 6)$$

$$= -16 - 36 - 64$$

$$= -116$$

And,

$$\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2} = \sqrt{(-6 + 2)^2 + (1 + 7)^2 + (-14 + 6)^2}$$

$$\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2} = 2\sqrt{29}$$

Putting all the values, we get

$$d = \frac{-116}{2\sqrt{29}}$$

$$d = \frac{-58}{\sqrt{29}} \Rightarrow \frac{-58\sqrt{29}}{29}$$

$$d = \frac{-58}{\sqrt{29}} \Rightarrow |d| = 2\sqrt{29}$$

Therefore, the distance between the lines is $2\sqrt{29}$ units.

14. Find the shortest distance between the lines whose vector equations are $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(\hat{i} - 3\hat{j} + 2\hat{k})$ and $\vec{r} = 4\hat{i} + 5\hat{j} + 6\hat{k} + \mu(2\hat{i} + 3\hat{j} + \hat{k})$

Ans: We have been given lines $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(\hat{i} - 3\hat{j} + 2\hat{k})$ and $\vec{r} = 4\hat{i} + 5\hat{j} + 6\hat{k} + \mu(2\hat{i} + 3\hat{j} + \hat{k})$

As we know that the shortest distance between the lines can be found by,

$$d = \frac{|\left(\vec{b}_1 \times \vec{b}_2\right)(\vec{a}_2 - \vec{a}_1)|}{|\vec{b}_1 \times \vec{b}_2|}$$

Now, from the given lines, we got

$$\vec{a}_1 = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b}_1 = \hat{i} - 3\hat{j} + 2\hat{k}$$

$$\vec{a}_2 = 4\hat{i} + 5\hat{j} + 6\hat{k}, \vec{b}_2 = 2\hat{i} + 3\hat{j} + \hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = (4\hat{i} + 5\hat{j} + 6\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= 3\hat{i} + 3\hat{j} + 3\hat{k}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 2 \\ 2 & 3 & 1 \end{vmatrix} \Rightarrow \vec{b}_1 \times \vec{b}_2 = -9\hat{i} + 3\hat{j} + 9\hat{k}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{(-9)^2 + 3^2 + 9^2} = 3\sqrt{19}$$

Now, putting all the values, we get

$$d = \left| \frac{9}{3\sqrt{19}} \right| = \frac{3}{\sqrt{19}}$$

Therefore, the shortest distance between the lines is $\frac{3}{\sqrt{19}}$ units.

15. Find the shortest distance between the lines whose vector equations are $\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$ and $\vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} - (2s+1)\hat{k}$.

Ans: We have been given lines $\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$ and $\vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} - (2s+1)\hat{k}$

$$\Rightarrow \vec{r} = \hat{i} - 2\hat{j} + 3\hat{k} + t(-\hat{i} + \hat{j} - 2\hat{k})$$

$$\text{and } \vec{r} = \hat{i} - \hat{j} + \hat{k} + s(\hat{i} + 2\hat{j} - 2\hat{k})$$

Now, the shortest distance can be found by,

$$d = \left| \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

Now, from the given lines we got,

$$\vec{a}_1 = \hat{i} - 2\hat{j} + 3\hat{k}, \vec{b}_1 = -\hat{i} + \hat{j} - 2\hat{k}$$

,

$$\vec{a}_2 = \hat{i} - \hat{j} - \hat{k}, \vec{b}_2 = \hat{i} + 2\hat{j} - 2\hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = (\hat{i} - \hat{j} - \hat{k}) - (\hat{i} - \hat{j} + \hat{k}) = -2\hat{k},$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & -2 \\ 1 & 2 & -2 \end{vmatrix}$$

$$\Rightarrow \vec{b}_1 \times \vec{b}_2 = 2\hat{i} - 4\hat{j} - 3\hat{k},$$

,

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{(2)^2 + (-4)^2 + (-3)^2} = \sqrt{29}$$

$$(\vec{b}_1 \times \vec{b}_2) \times (\vec{a}_2 \times \vec{a}_1) = (2\hat{i} - 4\hat{j} - 3\hat{k})(\hat{j} - 4\hat{k})$$

$$= -4 + 12$$

$$= 8$$

Putting all the values, we get

$$d = \left| \frac{8}{\sqrt{29}} \right| = \frac{8}{\sqrt{29}}$$

Therefore, the shortest distance between the lines is $\frac{8}{\sqrt{29}}$ units.

Miscellaneous Exercise

1. Find the angle between the lines whose direction ratios are a,b,c and b-c, c-a, a-b, .

Ans: As we know that, for any angle θ , with direction cosines, a,b,c and b-c, c-a, a-b can be found by,

$$\cos \theta = \left| \frac{a(b-c) + b(b-c) + c(c-a)}{\sqrt{a^2+b^2+c^2} \sqrt{(b-c)^2 + (c-a)^2 + (a-b)^2}} \right|$$

Solving this we get, $\cos \theta = 0$

$$\theta = \cos^{-1} 0$$

$$\Rightarrow \theta = 90^\circ$$

Therefore, the angle between the two lines will be 90° .

2. Find the equation of a line parallel to x-axis line passing through the origin.

Ans: As it is given that the line is passing through the origin and is also parallel to x-axis is x-axis,

Now,

Let us consider a point on x-axis be A

So, the coordinates of A will be $(a, 0, 0)$

Now, the direction ratios of OA will be,

$$\Rightarrow (a-0) = a, 0, 0$$

$$\text{The equation of OA} \Rightarrow \frac{x-0}{a} = \frac{y-0}{0} = \frac{z-0}{0} \Rightarrow \frac{x}{1} = \frac{y}{0} = \frac{z}{0} = a$$

Therefore, the equation of the line passing through origin and parallel to x-axis is $\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$.

3. if the lines $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-5}$ and $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$ are perpendicular Find the value of k

Ans: From the given equation we can say that $a_1 = -3, b_1 = 2k, c_1 = 2$ and $a_2 = 3k, b_2 = 1, c_2 = -5$.

We know that the two lines are perpendicular, if $a_1a_2 + b_1b_2 + c_1c_2 = 0$

$$-3(3k) + 2k \times 1 + 2(-5) = 0$$

$$\Rightarrow -9k + 2k - 10 = 0$$

$$\Rightarrow 7k = -10$$

$$\Rightarrow k = \frac{-10}{7}$$

Therefore, the value of k is $-\frac{10}{7}$

4. What is Find the shortest distance between these two lines

$$\vec{r} = 6\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(2\hat{i} - 2\hat{j} + 2\hat{k}) \text{ and } \vec{r} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$$

Ans: According to the question, we need to find the distance between the lines,

$$\vec{r} = 6\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(2\hat{i} - 2\hat{j} + 2\hat{k}) \text{ and } \vec{r} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$$

As we know we can find the shortest distance by,

$$d = \frac{|\left(\vec{b}_1 \times \vec{b}_2\right) \cdot (\vec{a}_1 - \vec{a}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

Now, from the equation of lines we get

$$\vec{a}_1 = 6\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\vec{b}_1 = \hat{i} - 2\hat{j} + 2\hat{k}$$

$$\vec{a}_2 = -4\hat{i} - \hat{k}$$

$$\vec{b}_2 = 3\hat{i} - 2\hat{j} - 2\hat{k}$$

$$\Rightarrow \vec{a}_2 \times \vec{a}_1 = (-4\hat{i} - \hat{k}) \times (6\hat{i} + 2\hat{j} + 2\hat{k}) = -10\hat{i} - 2\hat{j} - 3\hat{k}$$

$$\Rightarrow \vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix} = (4+4)\hat{i} - (-2-6)\hat{j} + (-2+6)\hat{k}$$

$$(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 \times \vec{a}_1) = (8\hat{i} + 8\hat{j} + 4\hat{k}) \cdot (-10\hat{i} - 2\hat{j} - 3\hat{k})$$

$$= -80 - 16 - 12$$

$$= -108$$

Now, putting these values in $d = \frac{|(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_1 - \vec{a}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$, we get

$$d = \frac{|-108|}{12} = 9$$

Therefore, the shortest distance between the above two lines is of 9 units.

5. Find the vector equation of the line passing through the points (1,2,-4) and perpendicular to the two lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$

Ans: According to the question, we get that $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{a} = \hat{i} + 2\hat{j} - 4\hat{k}$

We know that the equation of the line passing through point and also parallel to vector, we get

$$\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda (b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) \dots (1)$$

Now, the equation of the two lines will be

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7} \dots (2)$$

$$\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5} \dots (3)$$

As we know that line $\left(\text{line 1} \right)$ and $\left(\text{line 2} \right)$ are perpendicular to each other, we get

$$3b_1 - 16b_2 + 7b_3 = 0 \dots (4)$$

Also, we know that the line (1) and (3) are perpendicular to each other, we get

$$3b_1 + 8b_2 - 5b_3 = 0 \dots (5)$$

Now, from equation (4) and (5) we get that

$$\frac{b_1}{(-16)(-5) - 8(7)} = \frac{b_2}{7(3) - 3(-5)} = \frac{b_3}{3(8) - 3(-16)}$$

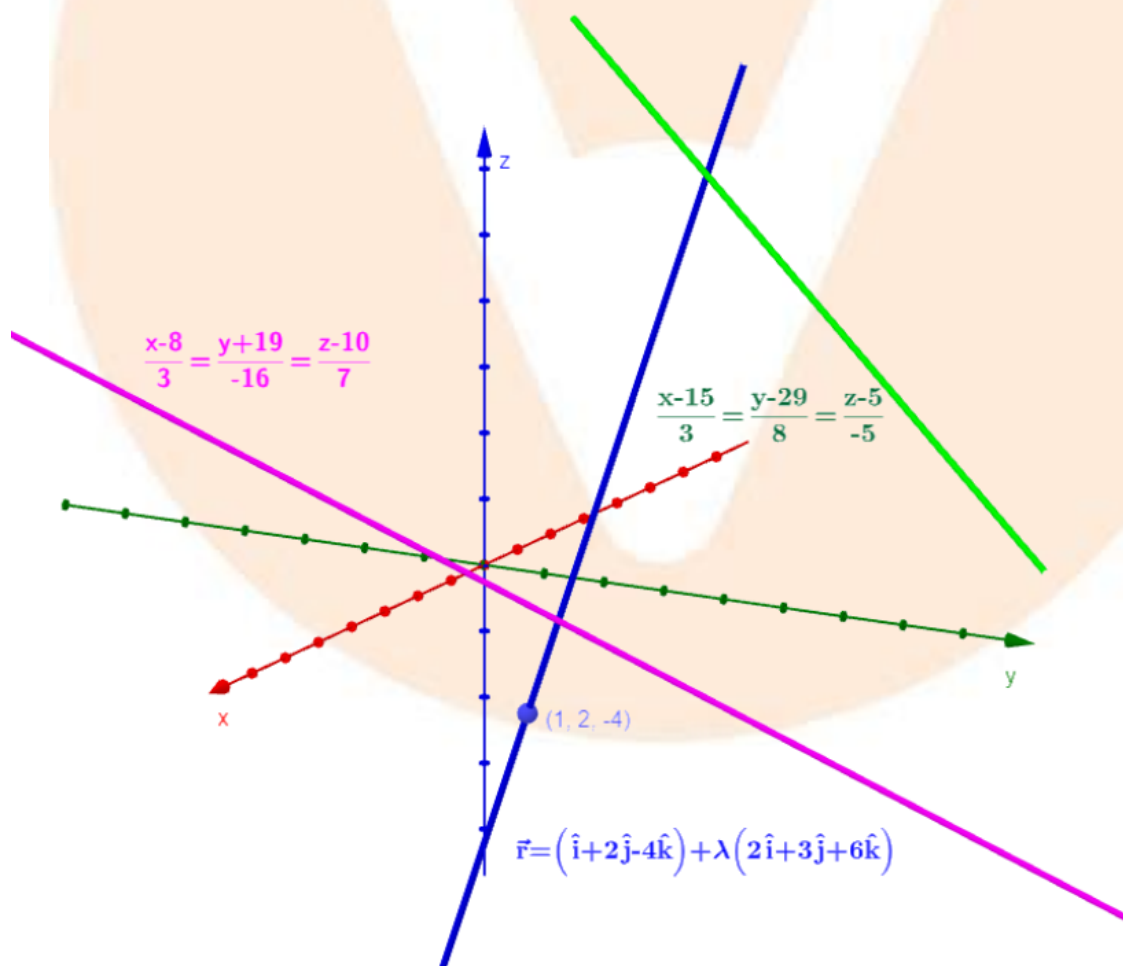
$$\Rightarrow \frac{b_1}{24} = \frac{b_2}{36} = \frac{b_3}{72} \Rightarrow \frac{b_1}{2} = \frac{b_2}{3} = \frac{b_3}{6}$$

Therefore, direction ratios of $\vec{b}$ are 2,3,6

Which means $\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$

Putting $\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ in equation (1), we get

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda (2\hat{i} + 3\hat{j} + 6\hat{k})$$



Therefore, the vector equation will be $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda (2\hat{i} + 3\hat{j} + 6\hat{k})$.

